

TOPIC 01: WHAT IS NODE.JS?

Definition:

Node.js ek **JavaScript Runtime Environment** hai jo JavaScript ko browser ke bahar server-side machine par run karne ki power deta hai.

CHROME V8 JS ENGINE

+

C++ BINDINGS & LIBUV

→

NODE.JS RUNTIME



Exam Catch: Node.js koi Programming Language, Library ya Framework **NAHI** hai! Yeh ek Runtime Environment hai.

Node REPL: Terminal me sirf node likhne se Read-Evaluate-Print-Loop start ho jata hai jahan direct JS execute kar sakte hain.

TOPIC 02: PROCESS & PROCESS.ARGV

What is process?

process ek global object hai jo current running Node.js process ki complete information aur control provide karta hai.

process.argv (Command Line Arguments):

Yeh ek array return karta hai jisme command-line se pass kiye gaye arguments store hote hain:

```
$ node app.js Shivam 22
```

Index	Value	Meaning
argv[0]	Path to Node.exe	Node runtime executable ka path
argv[1]	Path to app.js	Execute hone wali file ka path
argv[2]	"Shivam"	Pehla custom argument
argv[3]	"22"	Doosra custom argument

TOPIC 03: MODULE.EXPORTS & REQUIRE()

- `require()` → Ek built-in function hai jo alag file/module se exported code ko import karne ke kaam aata hai.
- `module.exports` → Ek special object hai jisme hum function, object ya value assign karke doosri files ke liye available banate hain.

MATH.JS: MODULE.EXPORTS = SUM; → `require('./math')` →

APP.JS: CONST SUM = REQUIRE(...)

Directory (Folder) Ko Require Karna:

Agar hum kisi pure folder ko require karte hain: `const fruits = require('./fruits');` toh Node.js automatically us folder ke andar maujood **index.js** file ko execute karta hai!



Analogy: Jaise Java/C++ me `main()` entry point hota hai, waise hi Node.js me kisi folder ka primary entry point **index.js** hota hai!

TOPIC 04: NPM (NODE PACKAGE MANAGER)

NPM Node.js ka standard default package manager hai. Iske 2 mukhya roop hain:

1. *LIBRARY / REGISTRY (ONLINE PACKAGES)*

2. *CLI TOOL (TERMINAL COMMANDS)*

```
$ npm install <package_name> # e.g. npm i express
```

TOPIC 05: PACKAGE FILES & FOLDERS

File / Folder	Kaam & Mahatva (Role)
node_modules/	Is folder ke andar aapke project ke saare installed packages aur unki sub-dependencies store hoti hain (heavy size, .gitignore me dalte hain).
package.json	Project ki metadata (name, version, scripts, direct dependencies list) store karta hai. Banane ke liye: <code>npm init -y</code> .
package-lock.json	Har ek installed dependency ka exact version number lock karke record karta hai taaki sabhi machines par identical setup bane.

TOPIC 06: LOCAL VS GLOBAL PACKAGES

- **Local Install:** `npm i <package>` → Sirf current project ke node_modules me install hota hai.
- **Global Install:** `npm i -g <package>` → Pure computer me globally install hota hai (CLI tools ke liye, e.g. nodemon).
- **npm link:** Globally installed package ko local project se connect/link karne ke liye use hota hai.

TOPIC 07: REQUIRE VS IMPORT

Feature	require() [CommonJS]	import [ES6 Modules]
Syntax	<code>const mod = require('./mod')</code>	<code>import { mod } from './mod.js'</code>
Selective Load	Puri file load karta hai (No selective)	Sirf zaroori functions load karta hai (Saves Memory! ✅)
Execution	Synchronous (blocking)	Asynchronous (non-blocking)
Setup	Default in Node.js	"type": "module" in package.json zaroori hai

